Openness and Growth: A Comparison of China and Mexico

Timothy Kehoe [1] and Xing Xu[2]

Nov. 2025

[1]University of Minnesota, Federal Reserve Bank of Minneapolis, and NBER
[2]University of Minnesota, Federal Reserve Bank of Minneapolis

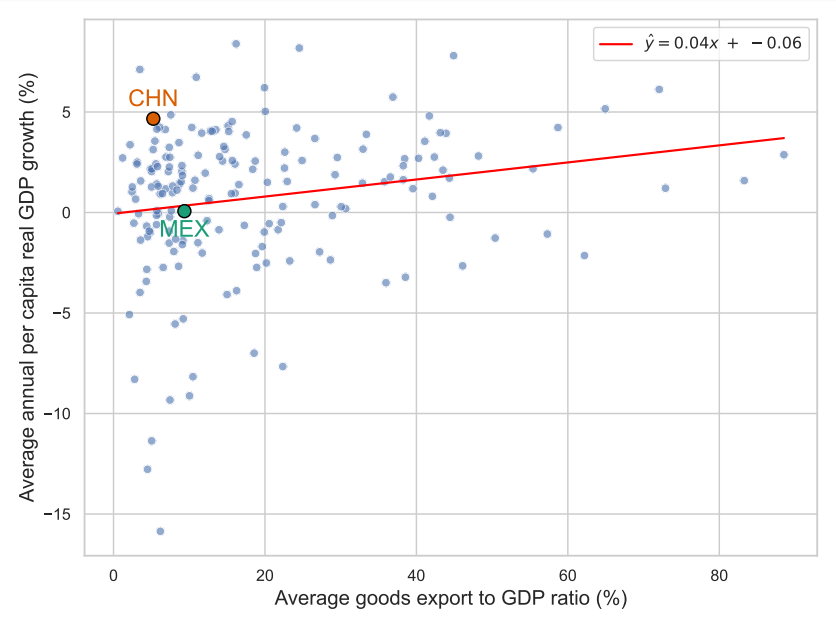
Does openness lead to growth?

- Canonical models link openness—trade and FDI—to faster growth. (Rivera-Batiz and Romer, 1991; Grossman and Helpman, 2018)
- Cross-country evidence finds little correlations between openness and growth.
- We revisit results of Kehoe and Ruhl (2010) and focus on China and Mexico; both opened to trade, Mexico started in late 80s, China in early 90s.

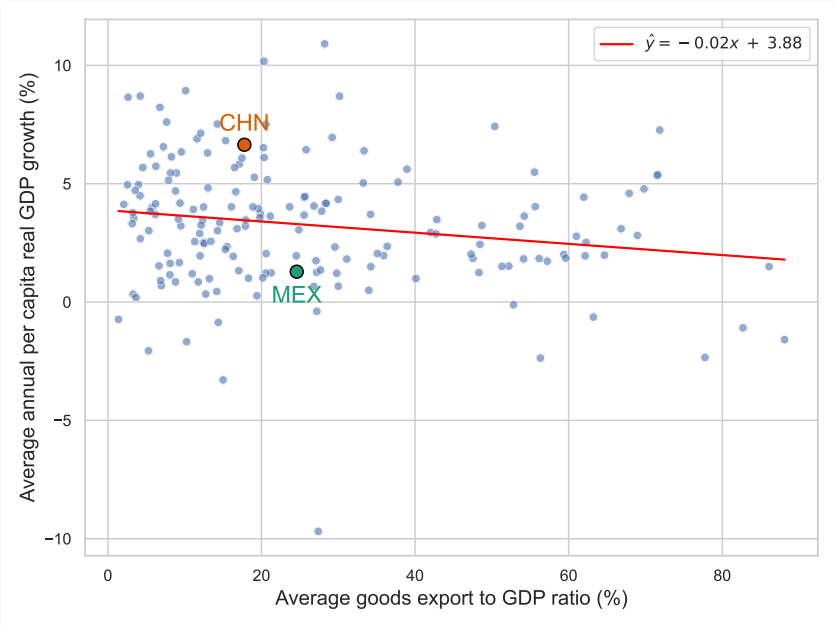
We propose and provide potential answers to two puzzles:

- Why did openness lead to growth in China but not in Mexico?
  Openness has to coincide with industrialization to generate growth.
- How did China sustain higher growth than Mexico despite less dependence on openness?
  Our hypotheses: moving up the value chain in China, infrastructure investment, and innovation incentives are key.

Trade openness and growth



(a) 1980-1999



(b) 2000-2019

Figure 1: Real per capita GDP growth rate vs export/GDP ratio

Key findings:

- Some but small positive correlation between openness and growth in 1980-1999.
- Negative correlation in 2000s!

Kehoe and Ruhl (2010)

- China's growth accelerated; Mexico's growth did not.
- Argued China was catching up to balanced growth path, but Mexico already had done so.
- **Argued Mexico did not grow due to inefficient financial system, lack of rule of law, and rigidities in labor market.**
- **Anticipated China growth would slow down due to similar problems.**

Updates:

- China's GDP per working age person caught up to Mexico's.
- China's openness fell after 2006.

China's GDP caught up with Mexico's

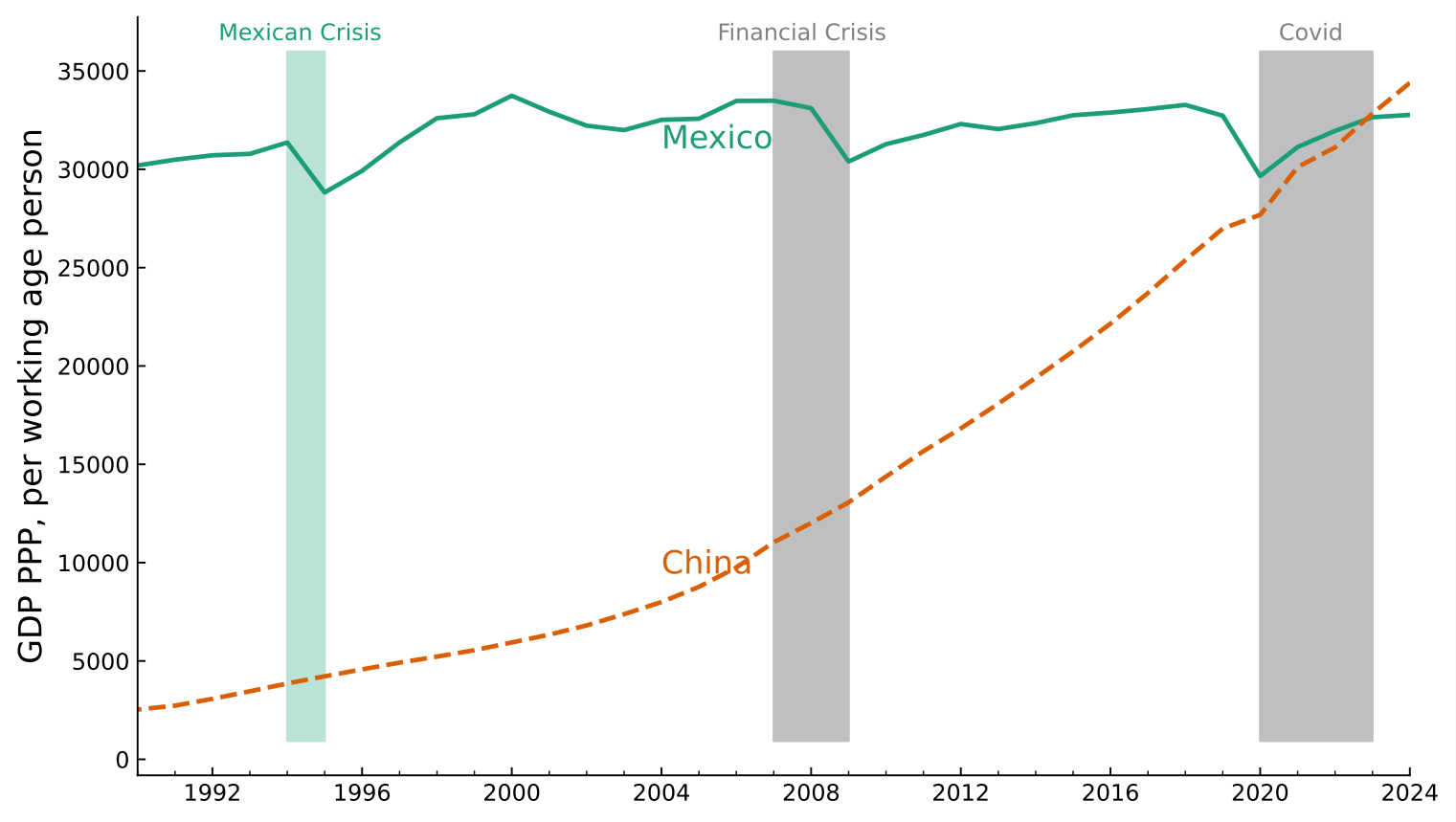


Figure 2: GDP per working age person in China and Mexico

Source: World Bank; PPP adjusted to 2021 USD

Mexico became more open than China

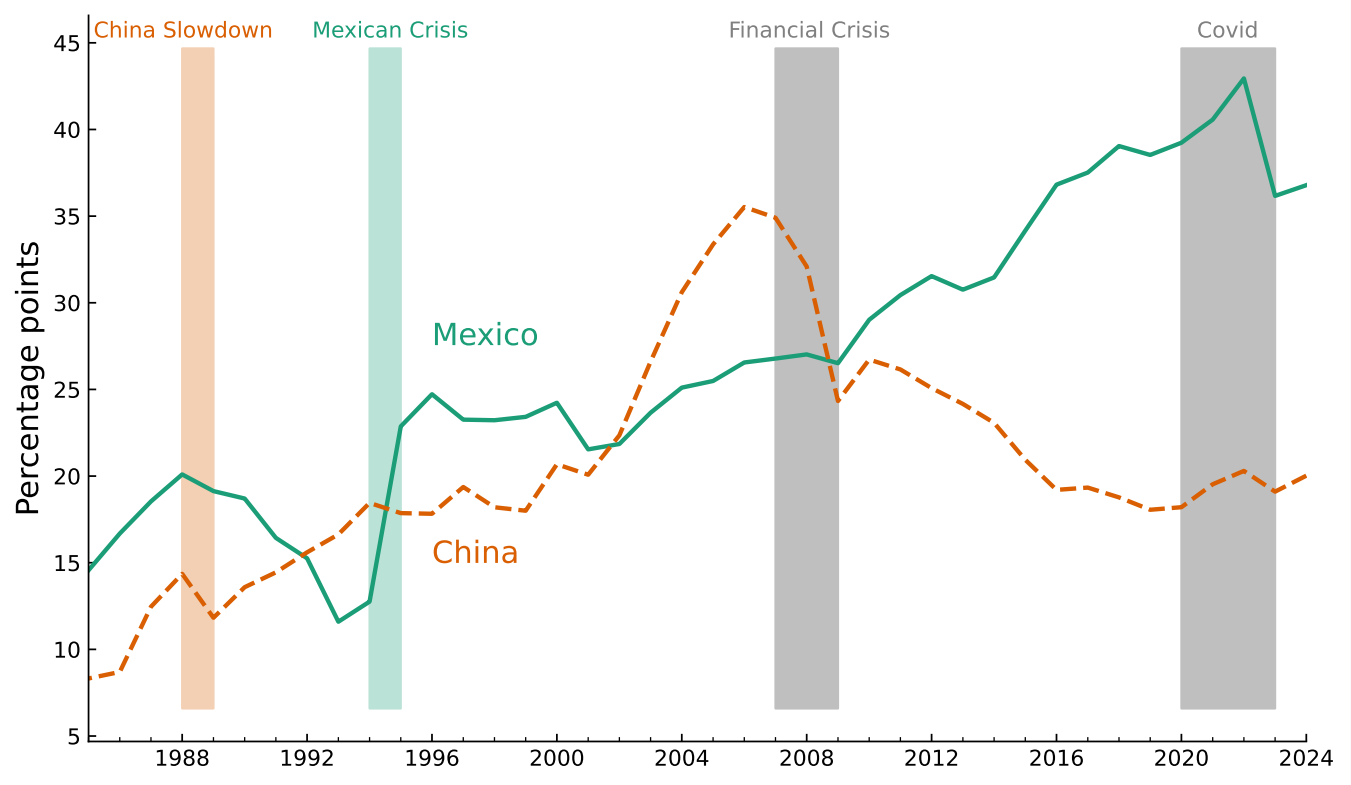


Figure 3: Export to GDP Ratio, % of GDP

Source: World Bank

FDI followed similar pattern

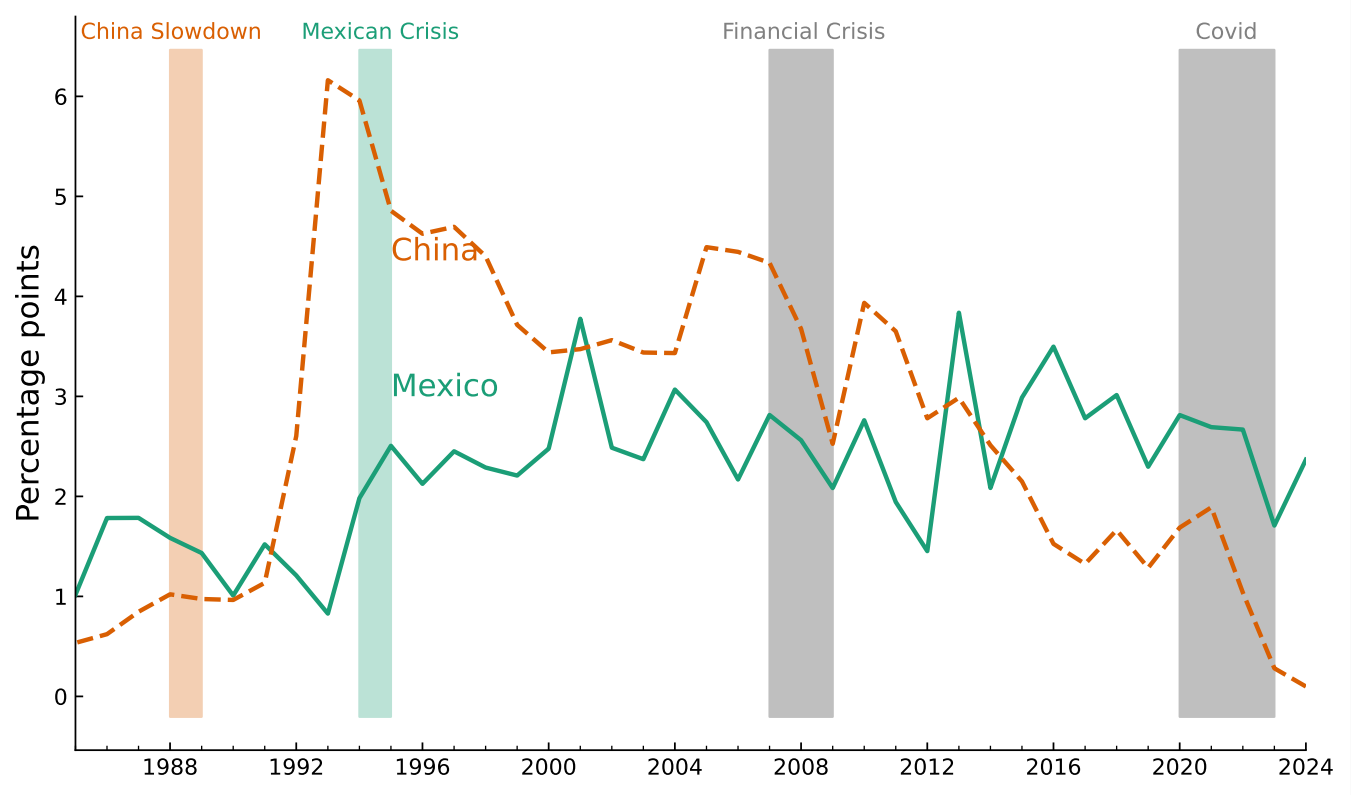
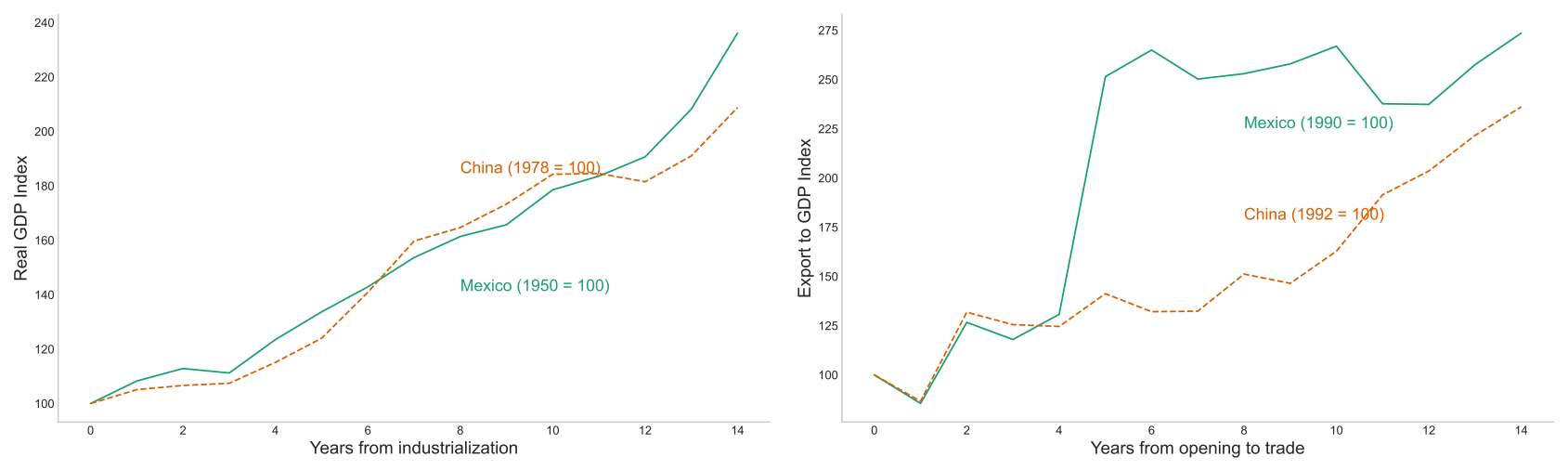


Figure 4: Foreign Direct Investment, net inflows, % of GDP

Source: World Bank.

First Puzzle: Why did China's openness lead to growth, but Mexico's did not?

Historical view



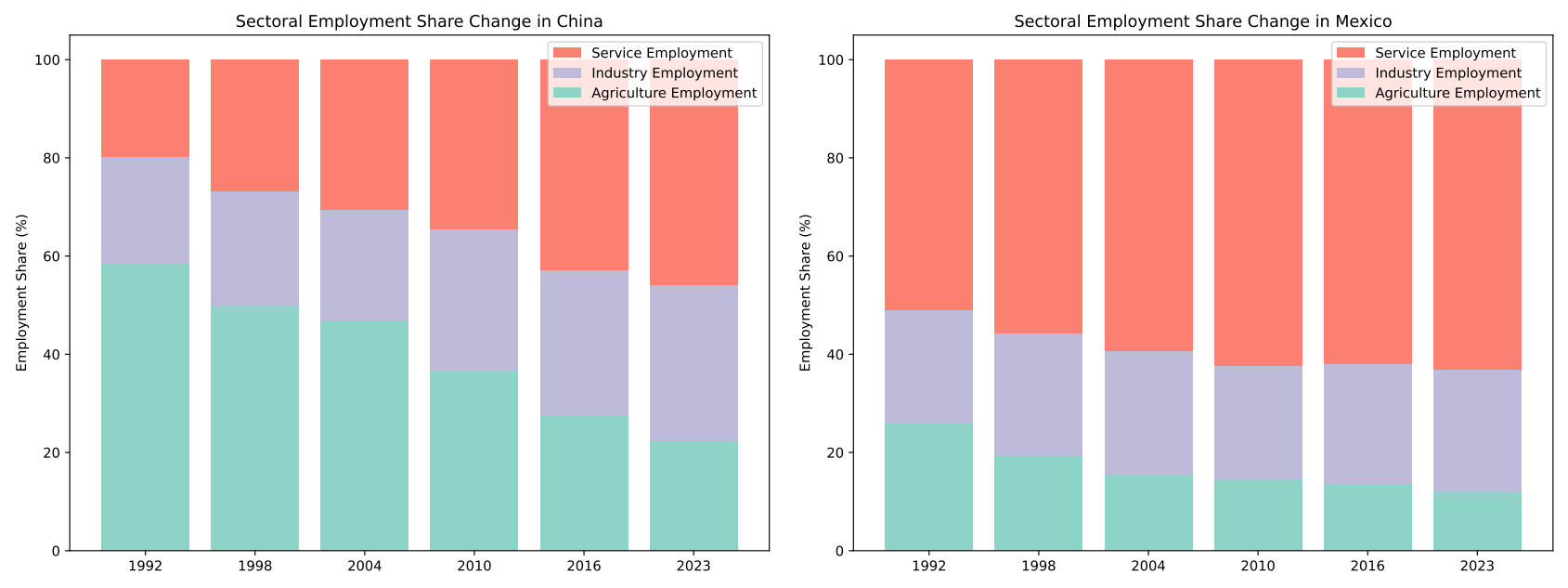
(a) GDP, PPP adjusted

(b) Export to GDP ratio

Figure 5: Industrialization and Trade Liberalization in China and Mexico
Source: World Bank; GDP PPP adjusted to 2015 USD

- Initial responses to industrialization and trade liberalization look similar.
- Timing is vastly different: 14 year gap for China, 40 for Mexico.
- Asturias et al. (2016) stresses importance of hysteresis of reforms.

China is still catching up in terms of structural transformation



(a) China **(b)** Mexico

Figure 6: Sectoral Employment Share Change, % of Total Employment

Source: International Labour Organization

Growth accounting: model

We follow Kehoe and Prescott (2007) and perform growth accounting using a Cobb-Douglas production function:

$$Y_t = A_t K_t^{\alpha} L_t^{1-\alpha},$$

rewrite it as:

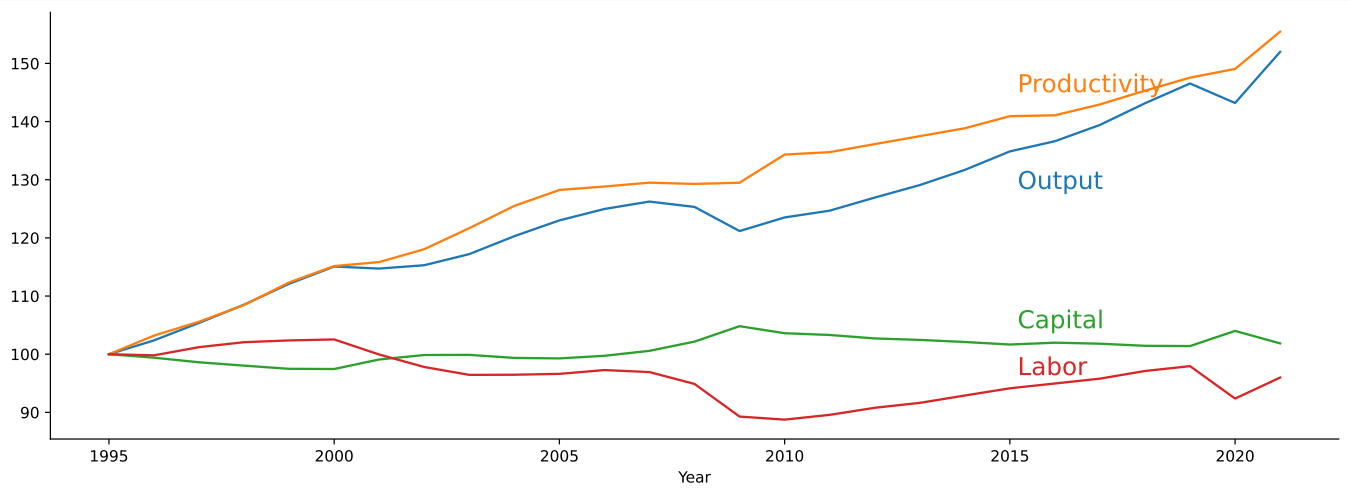
$$\frac{Y_t}{N_t} = A_t^{1/(1-\alpha)} \left(\frac{K_t}{Y_t}\right)^{\alpha/(1-\alpha)} \frac{L_t}{N_t}.$$

Components:

- Productivity: $A_t^{1/(1-\alpha)}$
- Capital deepening: $(K_t/Y_t)^{\alpha/(1-\alpha)}$
- Labor input per working-age person: L_t/N_t

Calibration uses 1990-2023 data; 1995 = 100; labor share fixed at 65%.

Growth accounting: United States

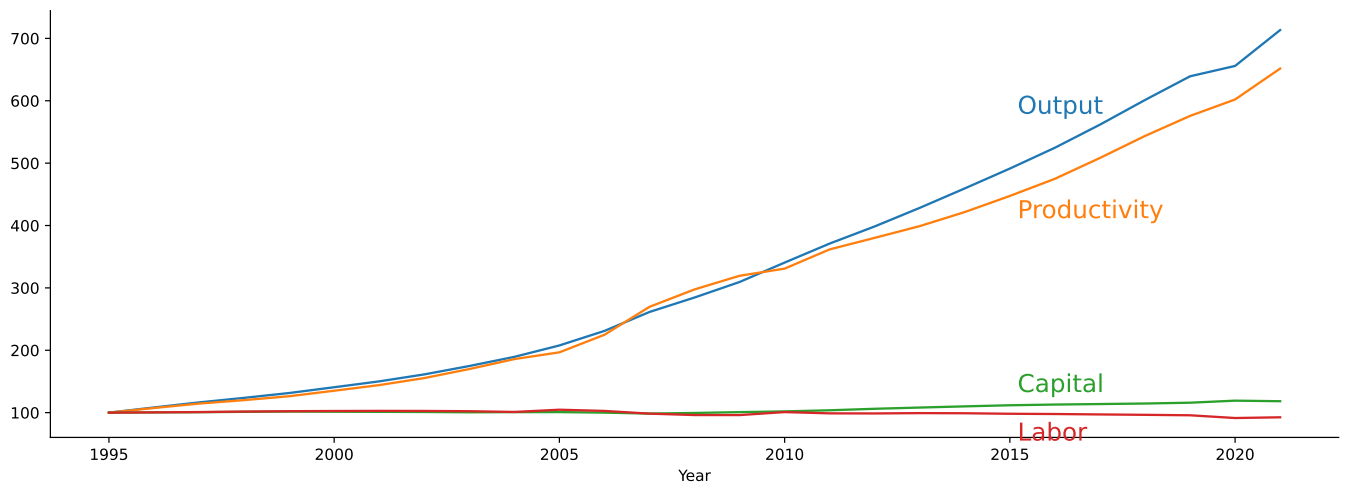


Source: World Bank, Conference Board, BEA, authors' computation

Key facts:

- Growth driven primarily by productivity.
- Capital-output ratio and hours per worker remain stable.
- Weak recovery after 2008 implies slower trend growth.

Growth accounting: China

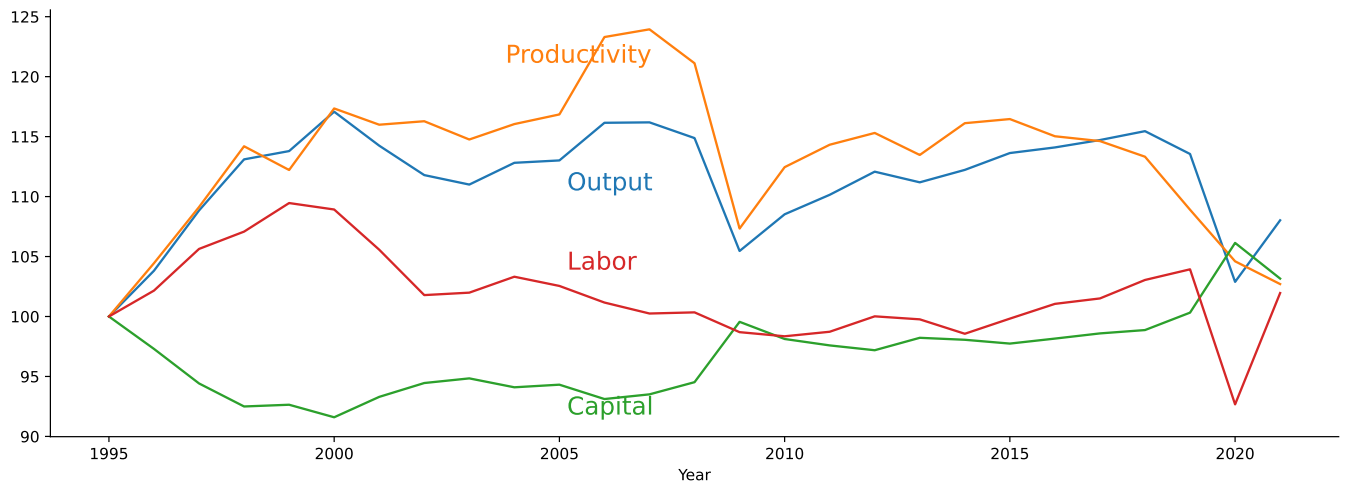


Source: World Bank, Conference Board, authors' computation

Key facts:

- Spectacular growth driven mainly by productivity.
- Capital deepening becomes more important after 2012.
- Barriers to riches (Kehoe and Ruhl (2010)) may bind as China approaches the frontier.

Growth accounting: Mexico



Source: World Bank, Conference Board, authors' computation

Key facts:

- Strong productivity-driven growth 1995–2005.
- Severe losses during the 2008–2012 crisis, little recovery after.
- Failure of productivity growth distinguishes Mexico from China and the U.S.

Going back to recent experience:

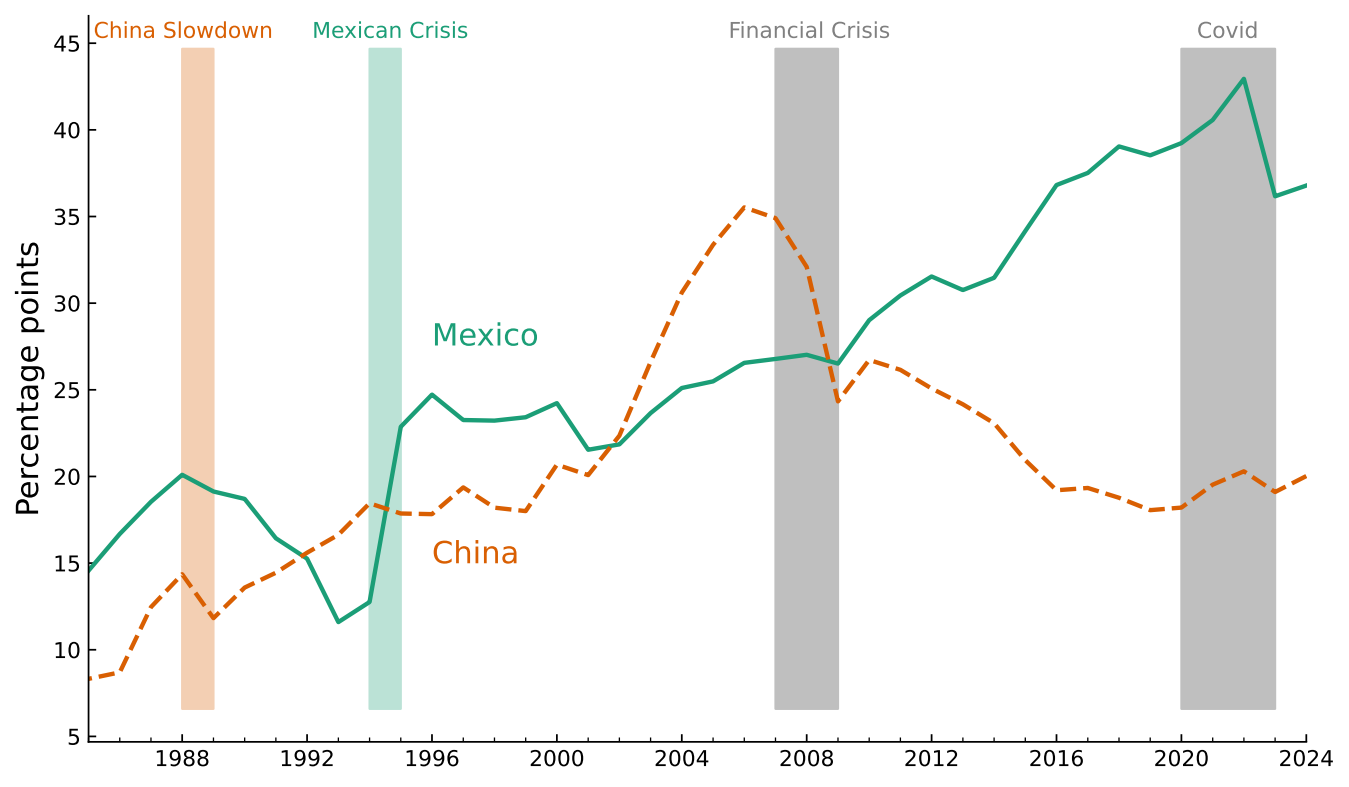


Figure 7: Export to GDP Ratio, % of GDP

Source: World Bank.

China's export/GDP ratio fell after 2006.

FDI followed similar pattern

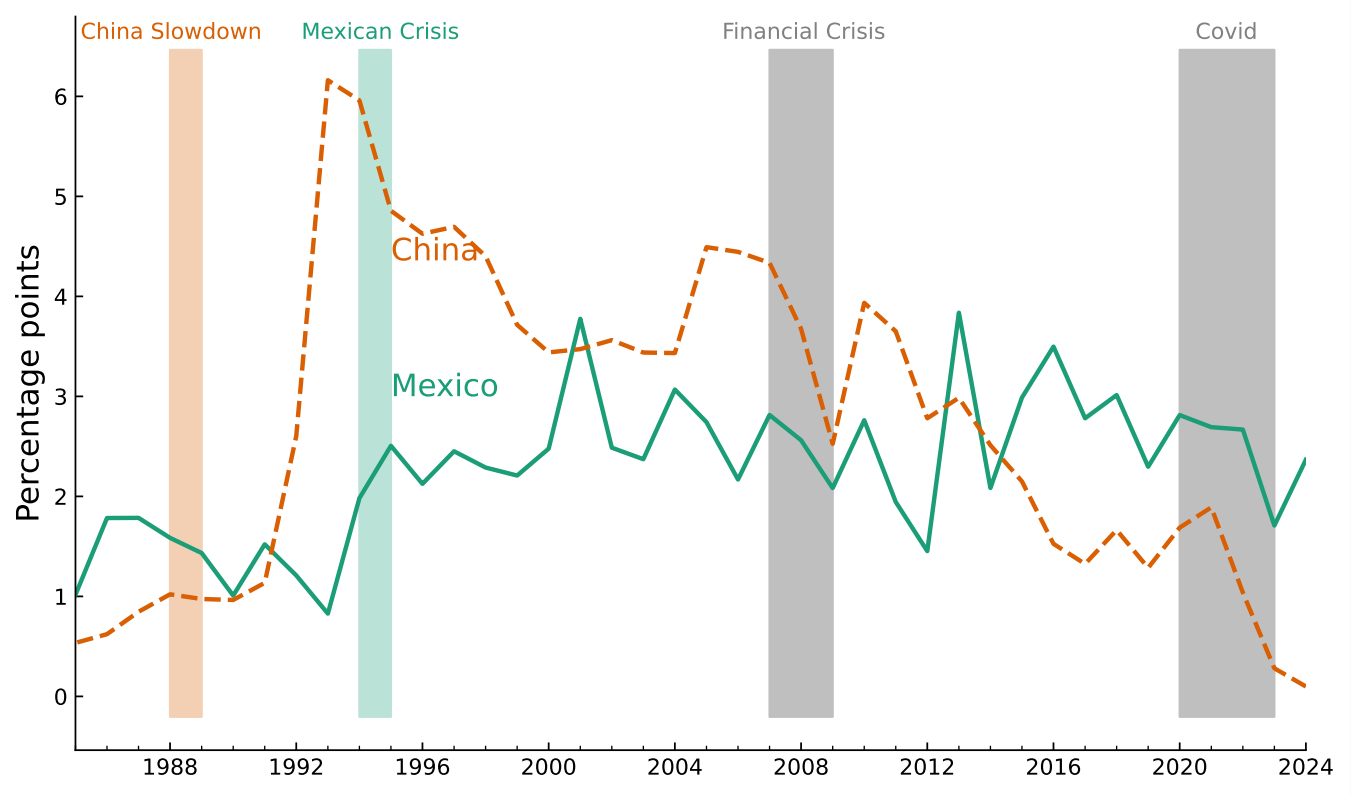


Figure 8: Foreign Direct Investment, net inflows, % of GDP

Source: World Bank.

Second Puzzle: How did China sustain growth despite less dependence on openness?

Hypothesis 1: China moving up the value chain

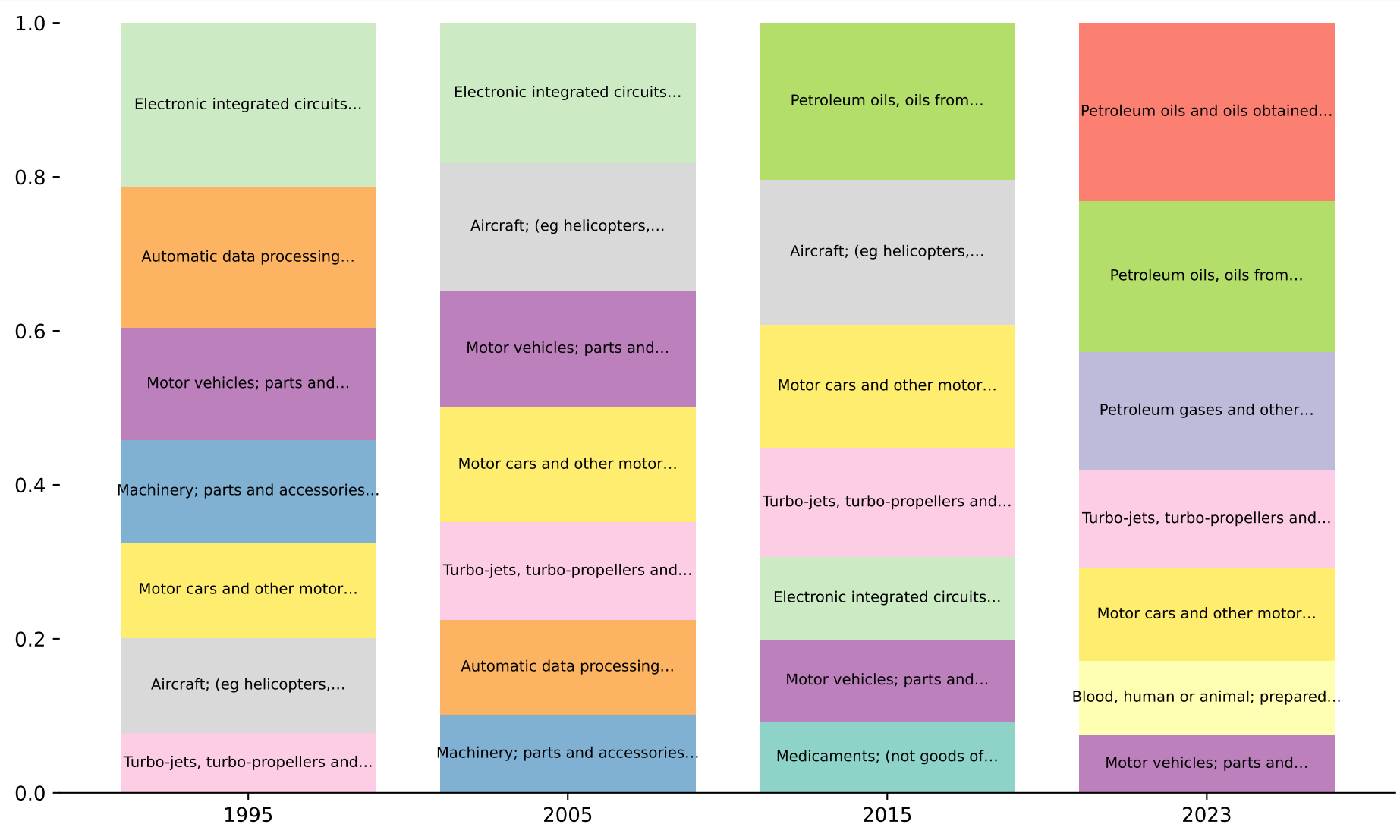
- By *value chain* we mean a ladder of products with increasing technological sophistication and quality (Young, 1991).
- Moving up the ladder—entering into high-value products and exiting low-value products—raises measured productivity and output.

Big picture:

- China: from textiles, toys, and simple consumer goods to solar panels, cars, semiconductors, smartphones, etc.
- Mexico: exports remain concentrated in assembled automobiles and petroleum products within the North American supply chain.

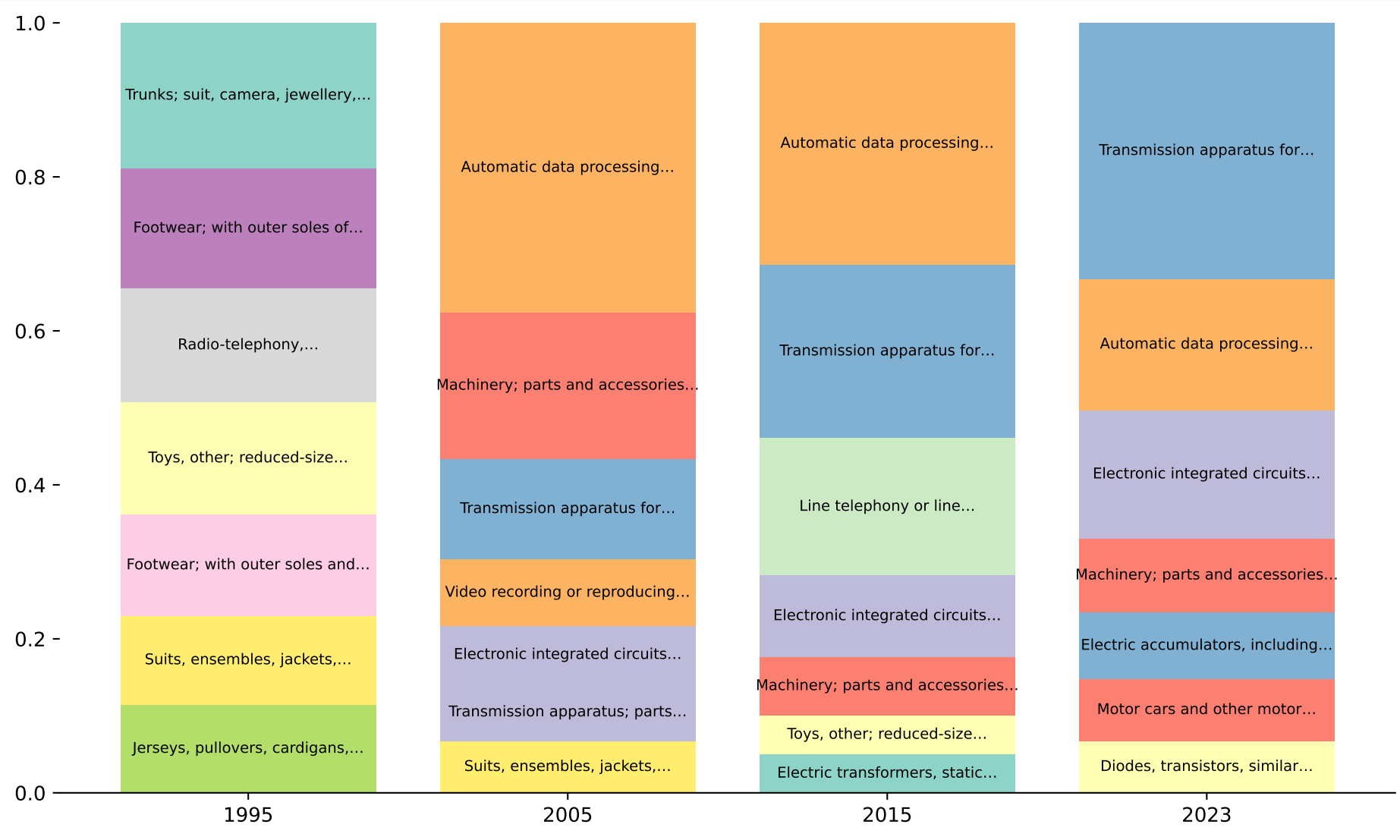
"Pobre México, tan lejos de Dios, tan cerca de Estados Unidos."
This phrase is often attributed to President Porfirio Diaz, but it is actually due to Mexican journalist and academic Nemesio García Naranjo.

U.S. export composition as benchmark



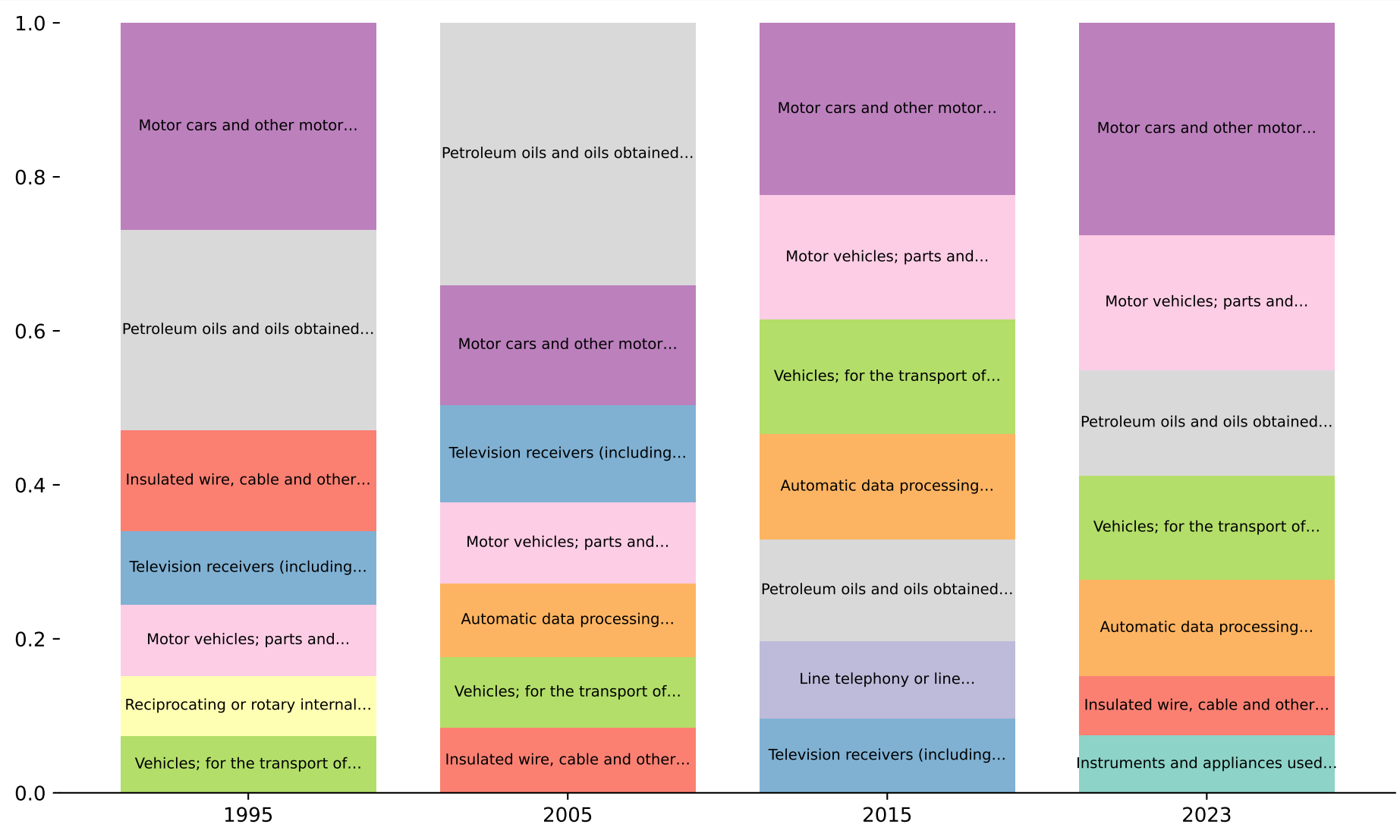
Source: CEPII BACI database.

China: moving into high-tech exports



Source: CEPII BACI database; exports from mainland to Hong Kong excluded.

Mexico: stuck in autos and oil



Source: CEPII BACI database.

U.S. Export destinations and diversification

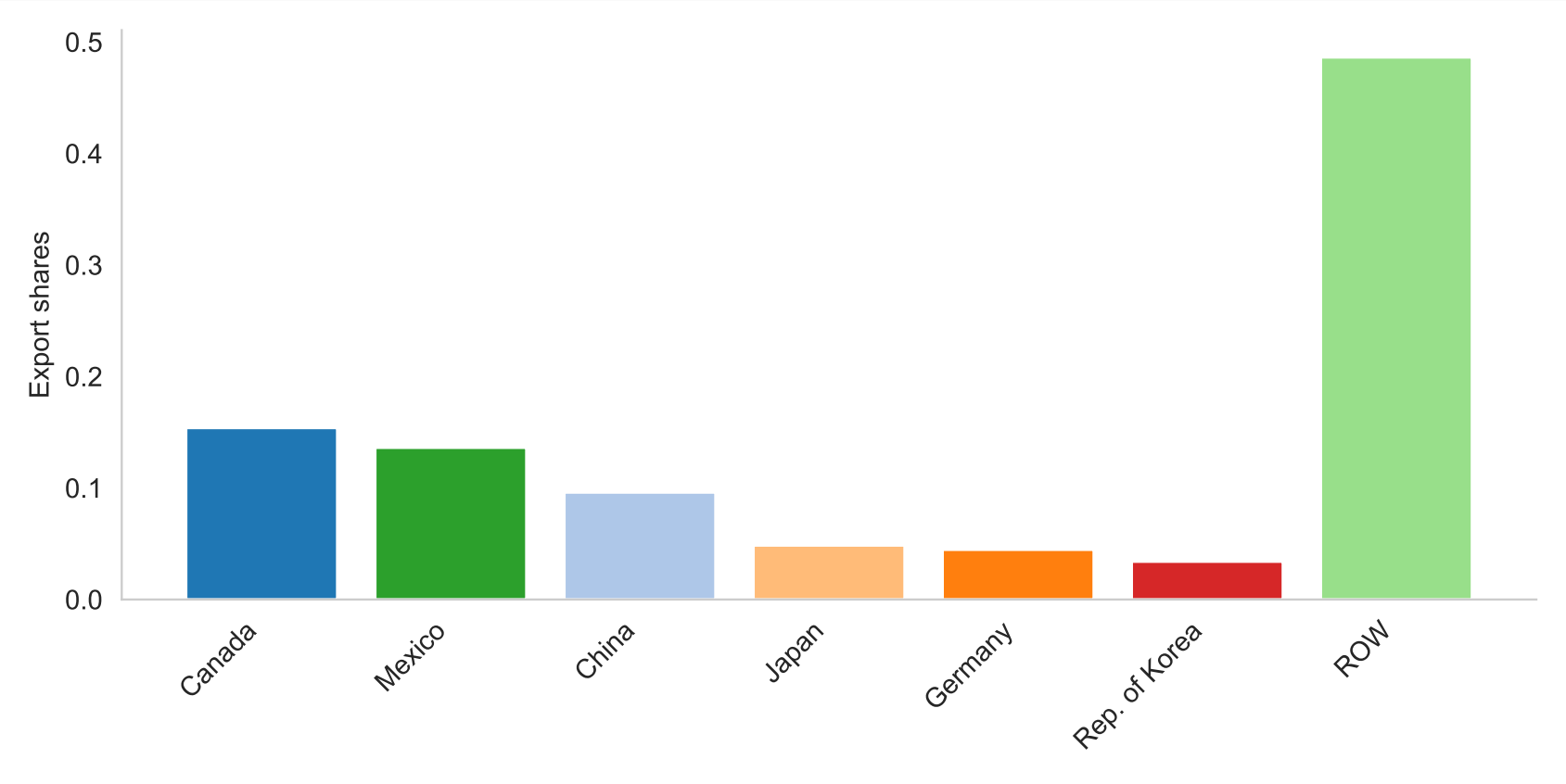


Figure 9: U.S. Export destinations in 2017

China's Export destinations and diversification

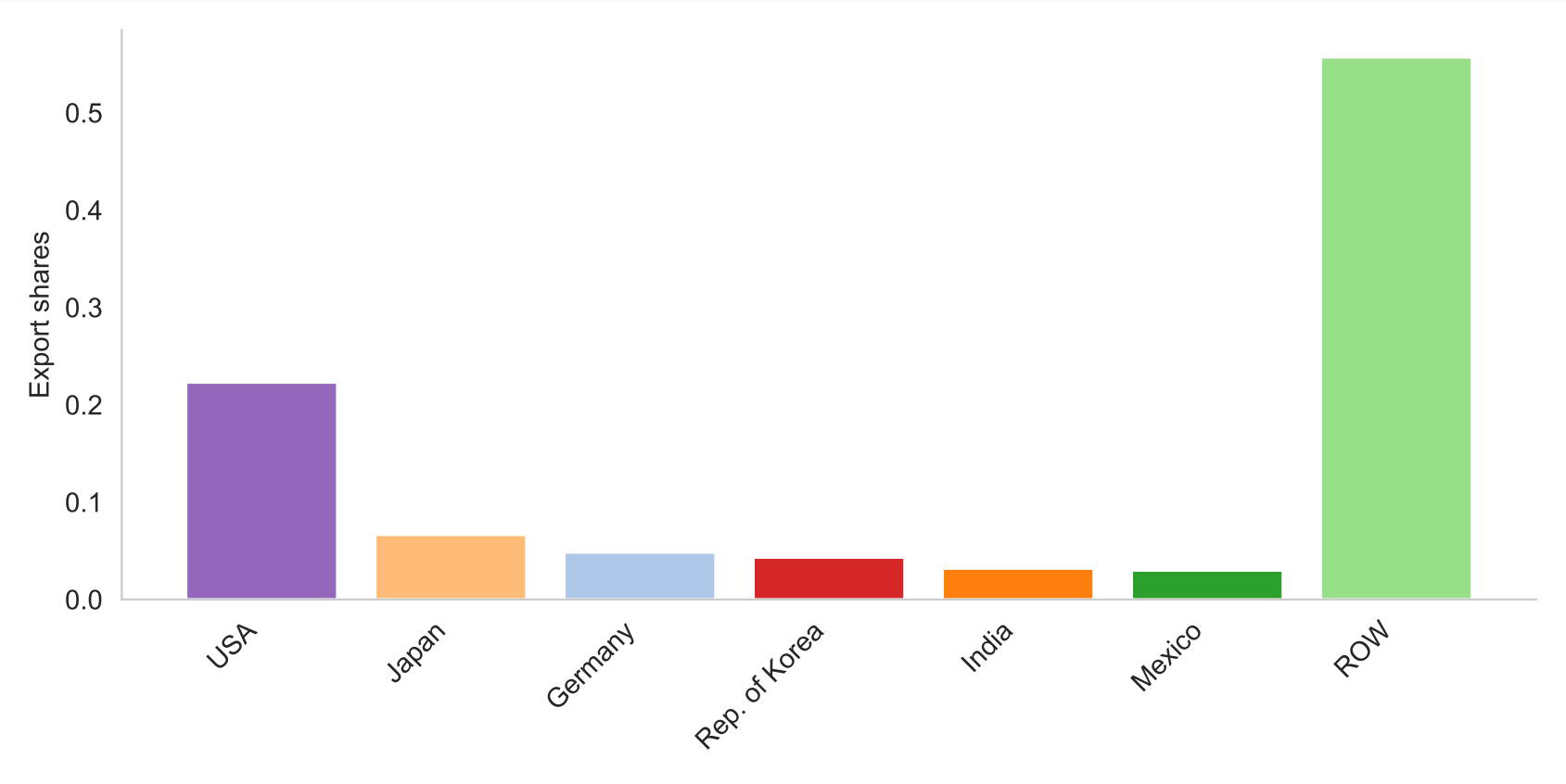


Figure 10: China's Export destinations in 2017

Mexico's Export destinations and diversification

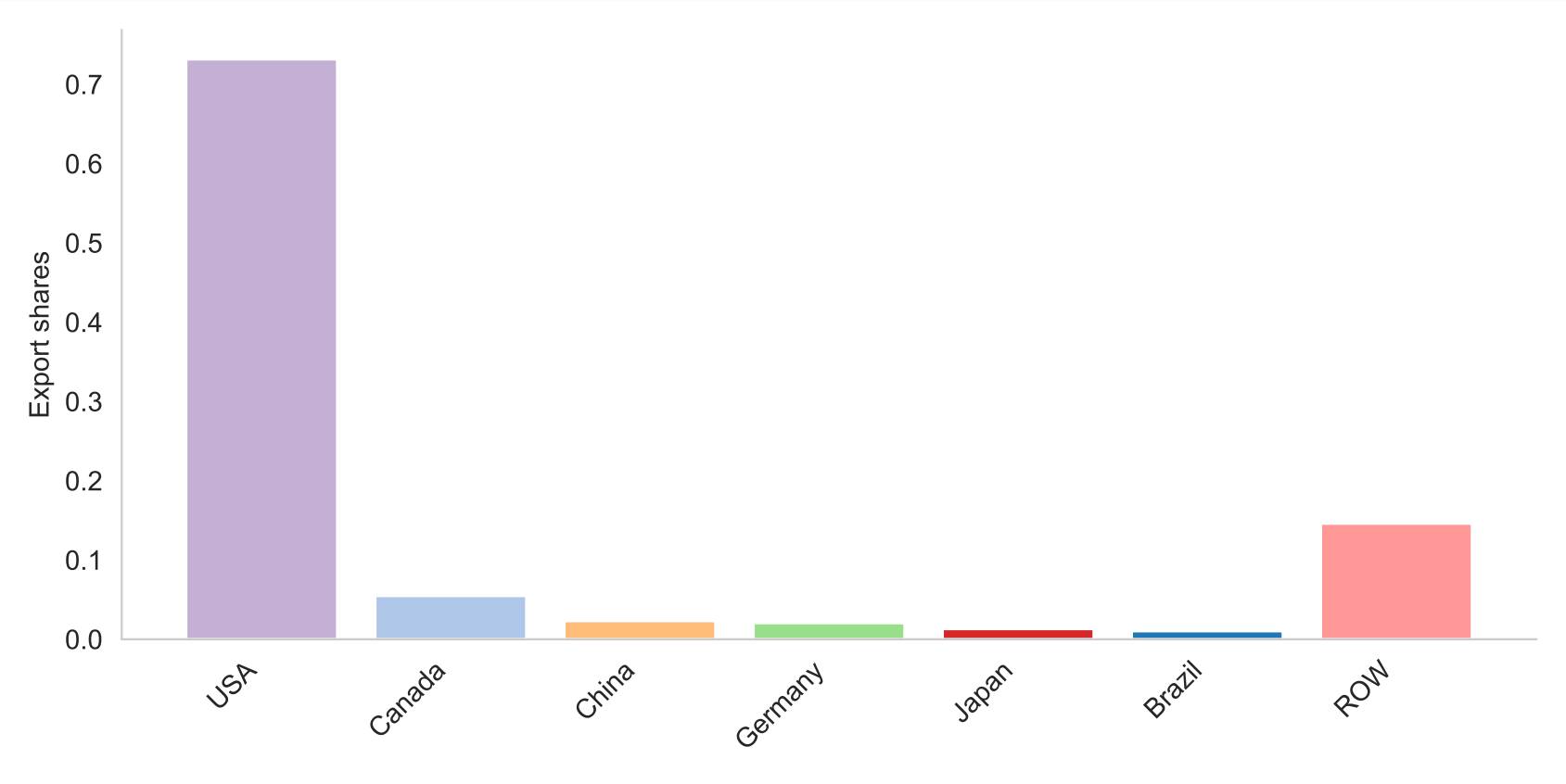


Figure 11: Mexico's Export destinations in 2017

Export destinations and diversification

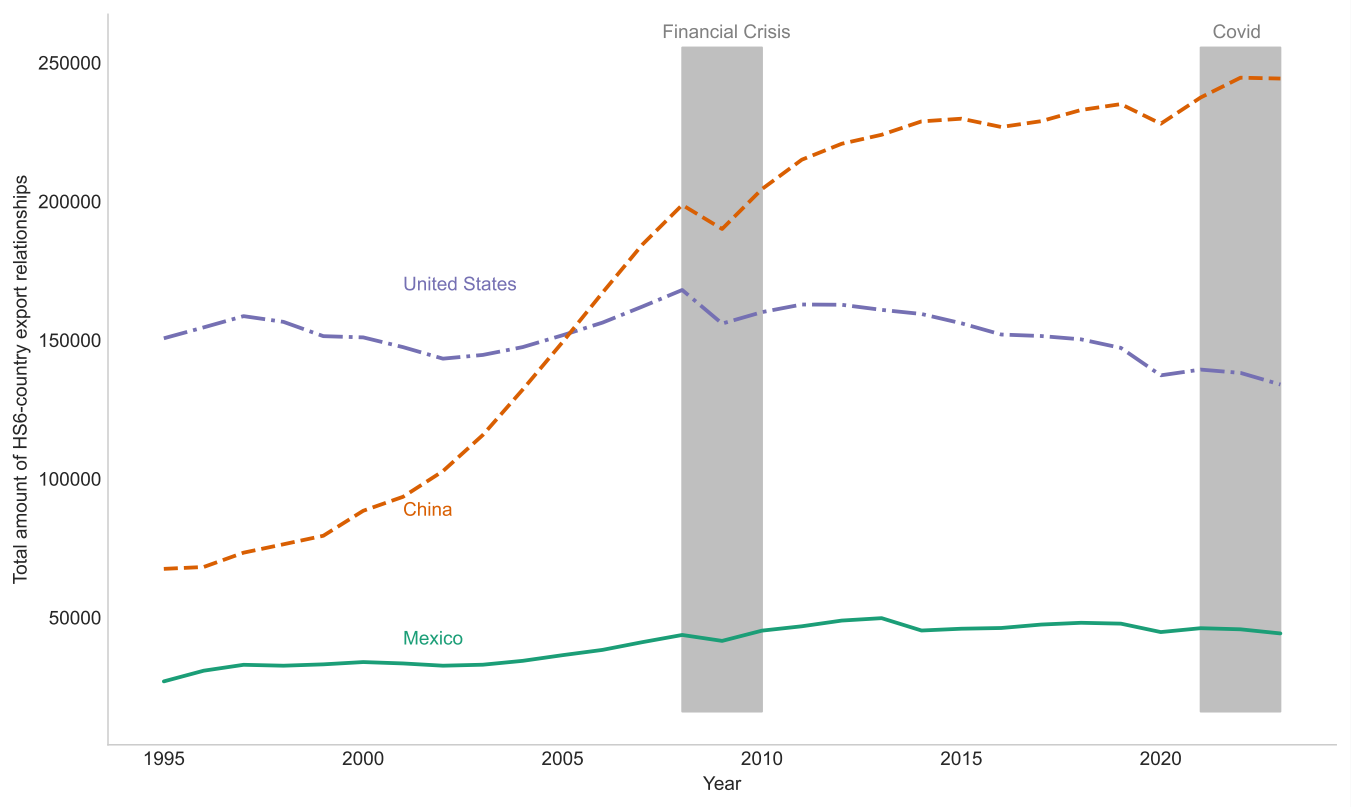
Partner concentration (2017):

- United States is largest export destination for both China and Mexico.
- U.S. share $\approx 20\%$ of China's exports vs. $\approx 70\%$ of Mexico's exports.
- United States and China have diversified export destinations; Mexico is highly concentrated on the U.S. market.

Implication:

- Mexico and Canada are the only large countries with exports so concentrated on a single partner.
- Reliance on the U.S. and role in the North American value chain may constrain Mexico's move up the value chain.

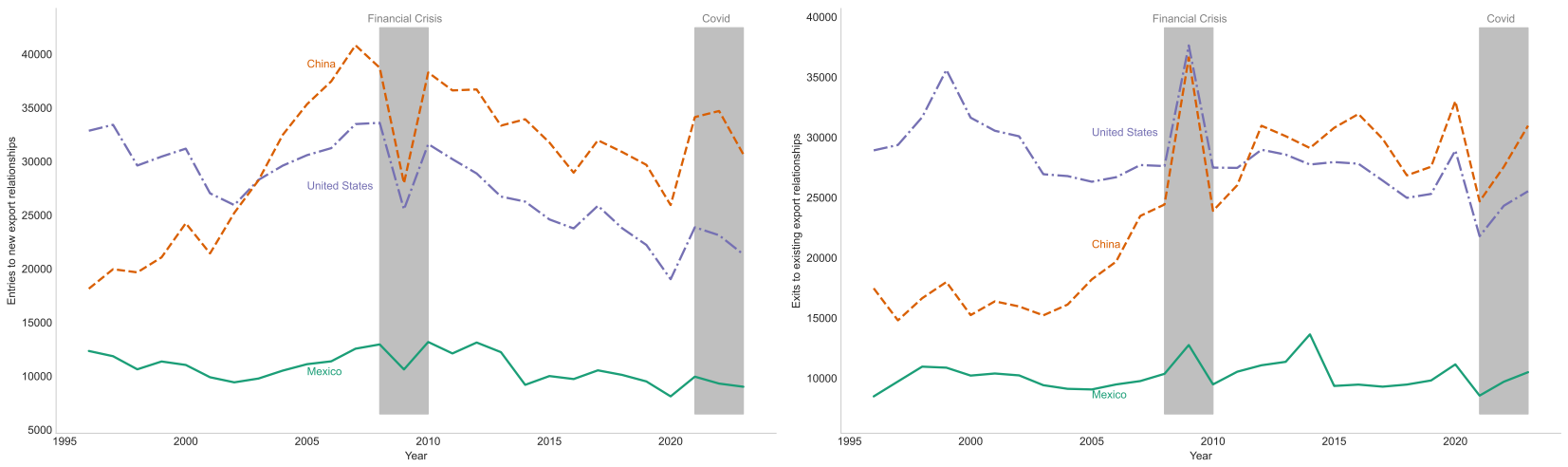
Extensive margin: total export relationships



Source: CEPII BACI database. Export relationships are country–HS6 pairs with export values above a threshold adjusted for inflation and export size; in 2023, the threshold is 1 million USD.

$$\text{Export relationship}_{it} \equiv \mathbb{1}\{\exp_{ijt}(\omega) \geq \text{Threshold}_{it}\}$$

Extensive margin: entries and exits



(a) Entries **(b)** Exits

Source: CEPII BACI database. Export relationships are country–HS6 pairs with export values above a threshold; the threshold is 1 million USD in 2023, adjusted for inflation and export size.

$$\text{Export relationship}_{it} \equiv \mathbb{1}\{\exp_{ijt}(\omega) \geq \text{Threshold}_{it}\}$$

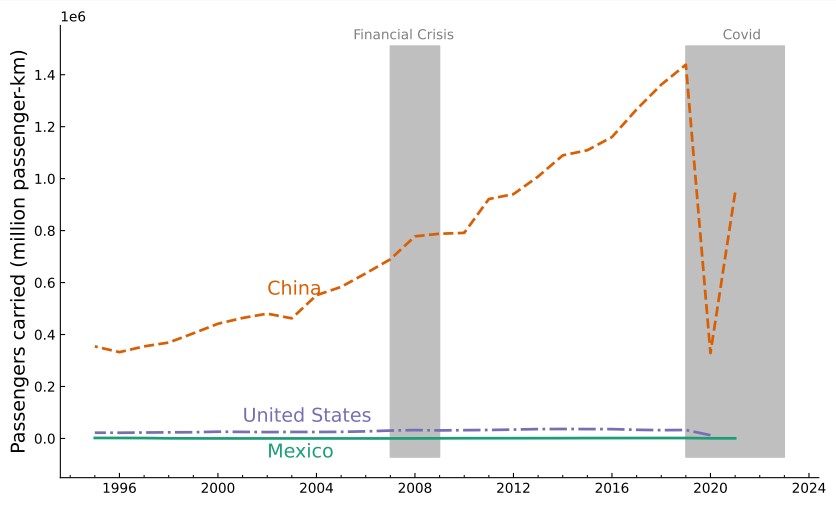
Moving up the value chain: takeaways

- China moved up the value chain: shifted export mix toward high-tech products, diversifies markets, and expanded/churned at the extensive margin.
- Mexico remained concentrated in autos and petroleum, tightly linked to the U.S. market with limited product-market churning.
- These differences help explain why China's TFP growth has been stronger than Mexico's despite both being open economies.
- Moving up the value chain—not openness per se—appears central for turning trade into sustained productivity growth.

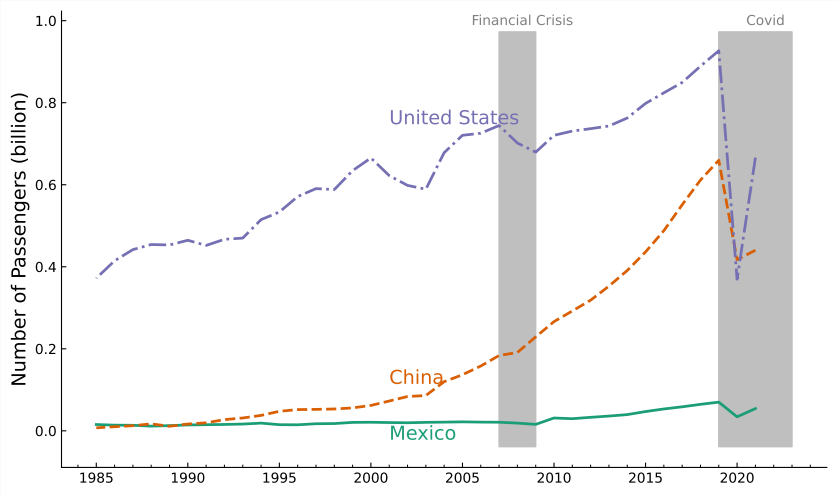
Hypothesis 2: more infrastructure investment

- China invests a far larger share of GDP in infrastructure.
- 2021: China $\approx$ 6.1% of GDP vs. Mexico $\approx$ 1.5%; upper-middle-income average $\approx$ 3.6%, by World Bank PPIAF.
- Xiong and Song (2024): China's infrastructure investment has been above 10% of GDP since 1998.
- Mexico's investment has remained persistently low.
- Both countries started with low infrastructure levels, but their paths diverged sharply.

China built much larger transportation network



(a) Distance Traveled by Railways



(b) Passengers by Air

Figure 13: Comparison of Passenger Transportation

Purple: US; Orange: China; Green: Mexico

Hypothesis 3: more R&D

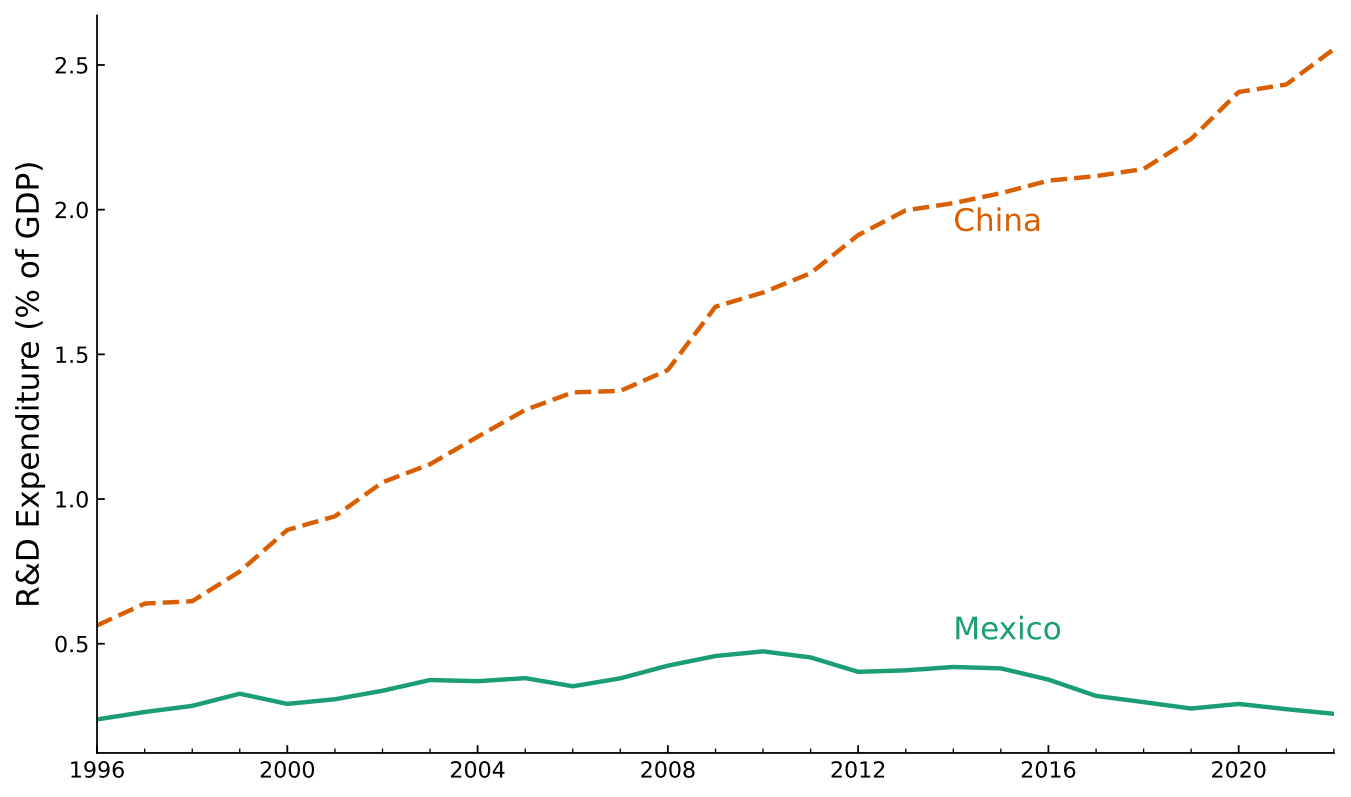


Figure 14: R&D expenditure, % of GDP

Source: UNESCO

China's patent applications surged with growth

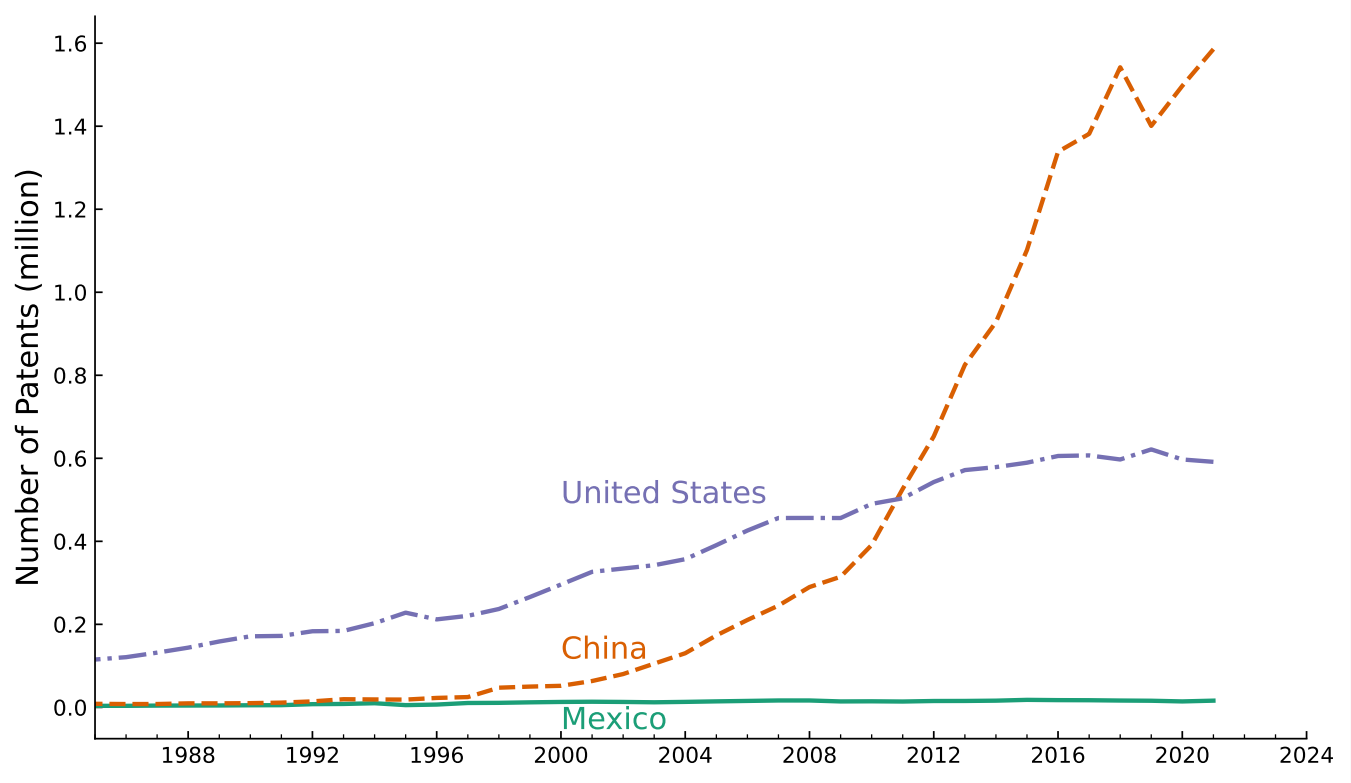
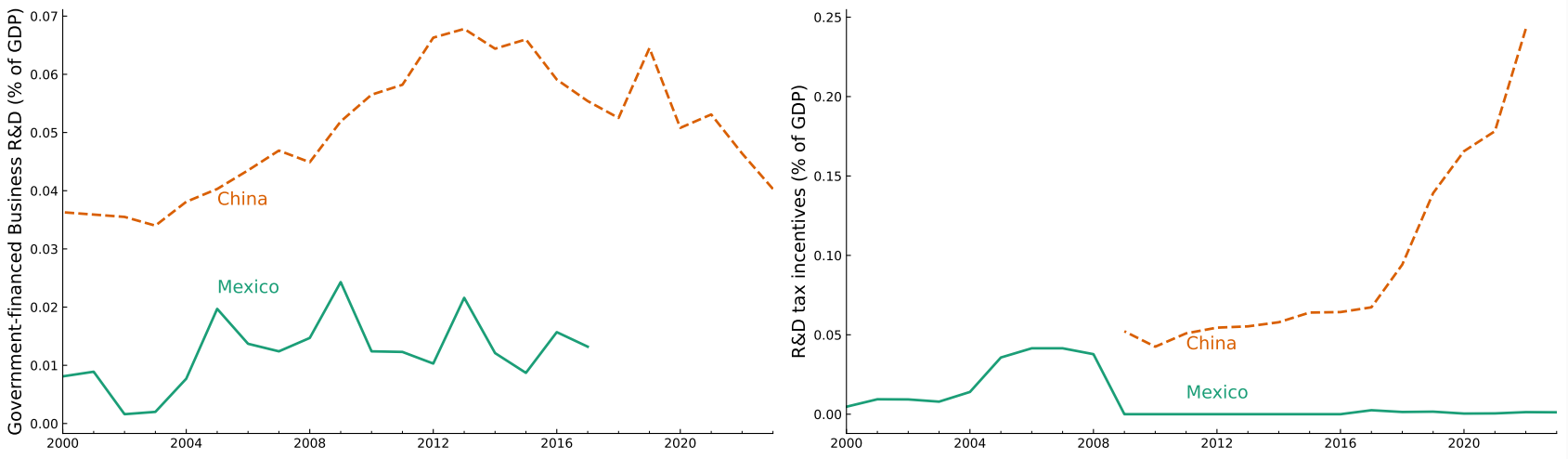


Figure 15: Total Number of Patent Applications

Source: World Intellectual Property Organization (WIPO)

Purple: US; Orange: China; Green: Mexico

China's industrial policy targeted firm innovation



(a) Government Financed Business R&D, % of GDP

(b) Government R&D Tax Incentives, % of GDP

Figure 16: Public Investment for Business R&D

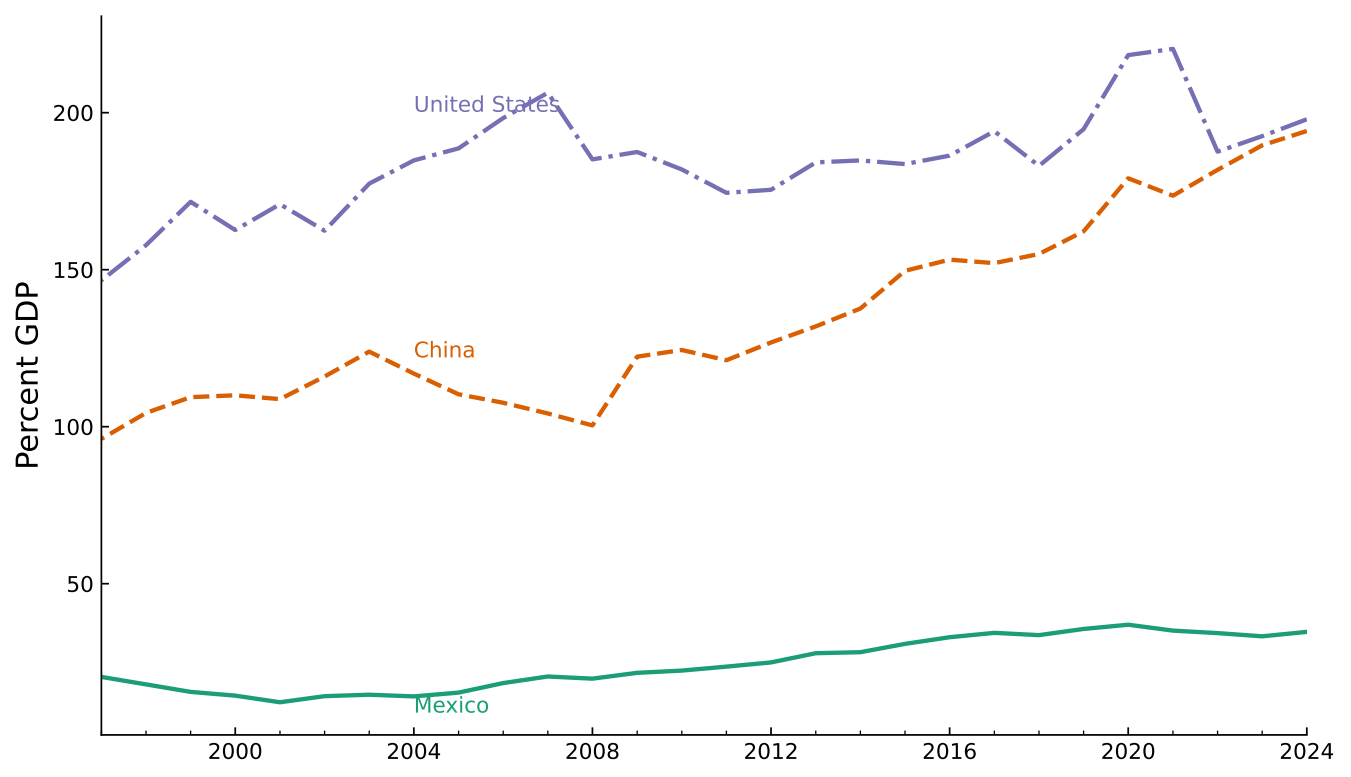
Source: OECD R&D tax expenditure dataset

Other industrial policies in China: *Reforms and privatizations of SOEs, Belt and Road Initiative, Made in China 2025, etc..*

Reflections on Kehoe and Ruhl (2010)

- **Instability in financial system:** has been more pronounced in China than Mexico (failure of Evergreen, pressure on local government bonds, closed financial markets).
- **Failure of rule of law:** more pronounced in Mexico than China (explosion of Drug Cartels since 2018).
- **Rigidities in labor market:** informality in Mexico remains high; China's Hukou system might have helped growth and is now gone (Wang and Conesa, 2022).

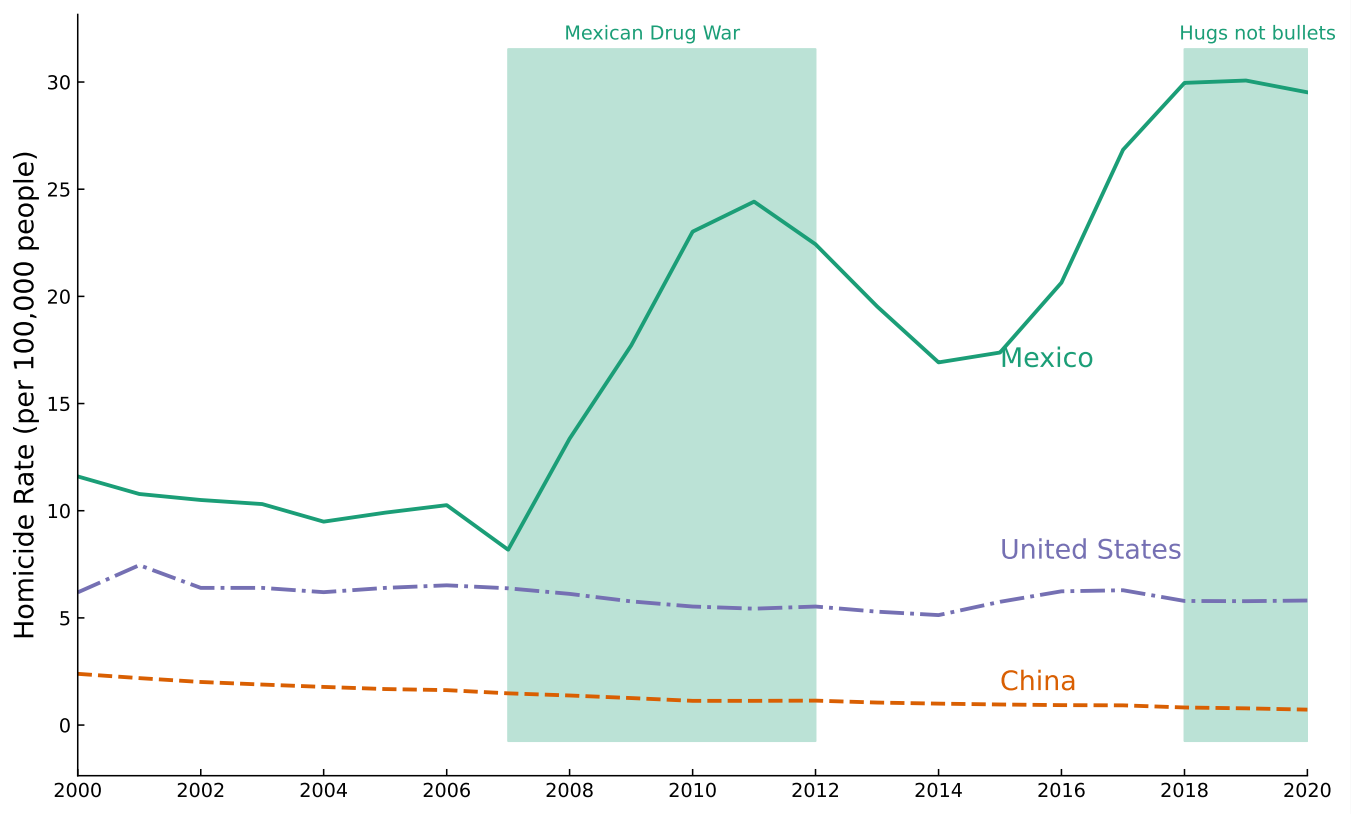
Barrier 1: Inefficient financial system



Source: World Bank

- Mexico's financial system remained stable during the GFC, but private credit provision is low.
- China liberalized rapidly since the 1990s; private credit expanded and approaches U.S. levels.

Barrier 2: Lack of rule of law



Source: WHO

- Homicide rate rose sharply after the 2006 drug war; little improvement after 2018.
- "abrazos, no balazos" by Lopez Abrador resulted in the cartels seizing control.

Barrier 2: Lack of rule of law (IP)

China:

- Major amendments to patent law (2008, 2020); stronger enforcement and penalties.
- IP courts improve adjudication; rapid rise in patent applications (see earlier figure).
- Enforcement uneven across regions; some local protectionism remains.

Mexico:

- Reforms enacted, but outcomes modest.
- 2024 International Property Rights Index: Mexico ranked 77/125; China 51/125.
- Weak IP protection constrains domestic innovation.

Barrier 3: Rigidities in the labor market

Mexico:

- Persistent dual labor market: rigid formal sector vs. large informal sector.
- Informality reduces tax base, weakens social insurance, and limits firm productivity growth.

China:

- Significant labor-market reforms, but Hukou and education frictions remain.
- Hukou historically restricted rural-urban migration; many restrictions eased over the past decade.
- Wang and Conesa (2022): early Hukou constraints may have aided growth by directing workers into productive urban sectors.

Overall: All three barriers remain relevant; without continued reforms, future growth may slow.

Conclusion

- It is hard to find evidence that openness to trade and FDI generates growth.
- Comparison of China with Mexico highlights some key confounding factors.
- Both countries opened to trade around same time; while openness in Mexico did not lead to growth, openness in China did, and China caught up with Mexico.
- Trade liberalization needs to coincide with, or follow closely, industrialization to generate sustained growth.
- We hypothesize domestic policies that promote moving up value chain, infrastructure, and innovation are key to transform openness into growth.

References

Asturias, J., Hur, S., Kehoe, T. J., and Ruhl, K. J. (2016). The Interaction and Sequencing of Policy Reforms. *Journal of Economic Dynamics and Control*, 72:45–66.

Grossman, G. M. and Helpman, E. (2018). Growth, Trade, and Inequality. *Econometrica*, 86(1):37–83.

Kehoe, T. J. and Prescott, E., editors (2007). *Great Depressions of the Twentieth Century*. Federal Reserve Bank of Minneapolis.

Kehoe, T. J. and Ruhl, K. J. (2010). Why Have Economic Reforms in Mexico Not Generated Growth? *Journal of Economic Literature*, 48(4):1005–1027.

Rivera-Batiz, L. A. and Romer, P. M. (1991). Economic Integration and Endogenous Growth. *Quarterly Journal of Economics*, 106(2):531–555.

Wang, Y. and Conesa, J. C. (2022). The Role of Demographics and Migration for the Future of Economic Growth in China. *European Economic Review*, 144(C):article 104076.

Xiong, W. and Song, Z. (2024). The Mandarin Model of Growth. https://wxiong.mycpanel.princeton.edu/papers/Mandarin.pdf.

Young, A. (1991). Learning by Doing and the Dynamic Effects of International Trade. *Quarterly Journal of Economics*, 106(2):369–405.